Suggested Teaching Guideline for
Fundamentals of Computer Networks PG-DITISS August 2025

Duration: 80 classroom hrs (40hrs theory + 40 lab hrs)

Objective: To introduce the student to fundamentals of computers and networks

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Data Communications and Networking, Behrouz A. Forouzan, McGraw Hill Education

Note: Every Session includes 2 Hours of theory and 2 Hours of Lab, unless indicated Otherwise.

Session 1 & 2:
- Internetworking
- OSI model
- Ethernet
- Wireless Networking

Assignment:
1) Difference between UTP & STP
2) Write categories of cables
3) e in CAT5e
4) OSI Model

Session 3:
- Internet Protocol
- TCP/IP model

Assignment:
1) IP
2) TCP/IP Model
3) Write a difference between TCP & UDP.

Session 4 & 5:
- IP Subnetting & variable Length Subnet Masking

Assignment:
1) Subnetting
2) Determine the network and host part of 192.168.5.85 /24, 10.128.240.50/30 address.
3) Divide the network as per following requirement 192.168.1.0/24
    a. Subnet 1 =28 hosts

b. Subnet 2 =52 hosts
c. Subnet 3 =15 hosts
d. Subnet4 = 5 hosts

Session 6:
- Router IOS & Security Device Manager

Assignment:
1) Subnet the Class C IP Address 195.1.1.0 So that you have 10 subnets each with a maximum 12 hosts on each subnet. List the Address on host 1 on subnet 0,1,2,3,10.
2) Divide the network to find 500 hosts in each subnet.
   152.152.0.0 /16

Session 7:
- Managing an Internetworking Router

Lab Assignments:
Working with Router Booting, configuration registers, Router IOS, Telnet, ResolvingHostname, Debugging.

Session 8:
- Static Routing
- Dynamic Routing
- Routing Protocols

Theory Assignment:
- Dynamic Routing
- Static Routing
- Routing Protocols

Session 9 & 10:
- Implementing of Routing Protocols

Lab Assignments:
- Implementation of Static Routing, RIP, IGRP, EIGRP, OSPF

Session 11:
- Layer 2 switching

Theory Assignment:
1) Spanning Tree Protocol (STP)?
2) Write types of Spanning Tree Protocol (STP).
3) priority number of Spanning Tree Protocol (STP)

4) disable Spanning Tree Protocol (STP)

Lab Assignments:

1. Configuration of switch, STP
2. Create the following topology for STP and configure a specific switch as root bridge by changing priority number.

Session 12:

- Virtual LANs

Lab Assignments:

1. Configure, verify, and troubleshoot VLANs (normal/extended range) spanning multipleswitches, inter-VLAN routing
2. Create following topology where Switch 1 is Server mode and switch 2,3,4,5 are client mode. Use network 192.168.10.0 /24
    - Transfer the Vlan info from server to all clients.
    - Assign vlan as mention in the topology.
    - PC's in having same vlan should ping each other.
3. Create following topology where Switch 1 is Server mode and switch 2,4, are client mode.
   Use network 192.168.1.0 /24
    - Transfer the Vlan info from server to all clients.
    - Assign vlan as mention in the topology.
    - PC's in having same vlan should ping each other.
    - Configured switch 3 in transfer mode and transfer vlan info to switch 4

Session 13:

- Infrastructure Security

Lab Assignments:

- Configure, verify, and troubleshoot port security, describe common access layer threat mitigation, Configure, verify, and troubleshoot IPv4 and IPv6 access list fortraffic filtering techniques,
- Verify ACLs using the APIC-EM Path Trace ACL analysis tool
- Configure, verify, and troubleshoot basic device hardening
- Describe device security using AAA with TACACS+ and RADIUS

Session 14:

- NAT
- IPV6
- WAN Technologies

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Lab Assignments
NAT configuration**,** Configuring Routers with IPv6

Session: 15

Lab Assignments:
- Configure and verify PPP and MLPPP on WAN interfaces using local authentication
- Configure, verify, and troubleshoot PPPoE client-side interfaces using localauthentication, Configure, verify, and troubleshoot GRE tunnel connectivity.
- Configure and verify single-homed branch connectivity using eBGP IPv4 (limitedto peering and route advertisement using Network command only)

Session 16:
- Introduction to SDN,
- Overview and Architecture of SDN
- Scalability (Data Centers, Service provider networks, ISP Automation)
- Reliability (QoS, and Service Availability)
- Consistency (Configuration management, and Access Control Violations)
- Opportunities and Challenges

Theory Assignment:
- SDN and Architecture of SDN.

Session 17:
- Virtual networking
- Use-cases (Network Access Control, Virtual Customer Edge, DatacenterOptimization)

Theory Assignment:
- Virtual networking
- Use cases of Network Access Control, Virtual Customer Edge, Datacenter Optimization

Session 18:
- Introduction to OpenFlow
- History and evolution
- Control and data plane separation
- OpenDaylight architecture overview

Lab Assignment:
- Understanding the classroom environment
- Installing OpenDaylight

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- Initial Configuration
- Initial Controller Interations

Session 19:
Getting Started with Opendaylight
- How to get started
- OpenDaylight clustering
- Model-Driven Service Abstraction Layer
- Internal datastore
- OpenFlow plugin
- OpenVSwitch concepts
- Mininet overview
- L2Switch Application

Lab Assignment:
- Configuring an OpenDaylight Cluster
- Initial Command Line Interactions
- Enabling Required Features
- Launching the User Interface
- Creating an Emulated SDN Network
- Running L2Switch

Session 20:
Getting more from Opendaylight
- OpenDaylight and AAA
- Introduction to OVSDB Virtualization
- Application Intents and Group Based Policy
- Service Function Chaining
- LISP Flow Mapping
- Virtual Tenant Networks

Lab Assignment:

- Advanced AAA integrations
- Creating Virtual Networks
- Working with policies and intents
- Initial Service Function Chaining
- OpenDaylight and OpenVSwitch